Parametric Hypothesis Tests — Part 3

Tests for Mean and Variance of Normal Populations

Paulo Fagandini

Lisbon Accounting and Business School – Polytechnic University of Lisbon

2025-04-21

Recap from Last Week

In Lecture 6, we established:

- The general framework: H_0 vs. H_1, test statistic, α, critical region
- Type I and Type II errors, power
- The p-value: smallest α at which we reject H_0
- The relationship between CIs and two-tailed tests

Today: We apply the framework systematically to every test case in the syllabus.

Road Map for Today

Section	Topic
3.3	Tests for μ (Normal, σ known and unknown)
	Tests for σ^2 (Normal population)
3.4	Tests for $\mu_1 - \mu_2$ (two Normal populations)
3.5	Large-sample (asymptotic) tests: mean and proportion

3.3 — Tests for Parameters of Normal Populations

Overview

We consider a random sample $(X_1, X_2, \ldots, X_n)$ from a **Normal population** $X \sim N(\mu, \sigma)$.

We will cover four cases:

- **Case 1**: Test for μ with σ **known**
- **Case 2**: Test for μ with σ **unknown** (small sample)
- **Case 2'**: Test for μ with σ **unknown** and $n > 30$
- **Case 3**: Test for σ^2 (or σ)

Case 1: Test for μ, σ Known

Test for μ — Normal population, σ known

Test statistic:

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \underset{\text{under } H_0}{\sim} N(0,1)$$

Critical regions (at significance level α):

H_1	Rejection region
$\mu \neq \mu_0$	$\|z_{obs}\| \geq z_{\alpha/2}$
$\mu > \mu_0$	$z_{obs} \geq z_\alpha$
$\mu < \mu_0$	$z_{obs} \leq -z_\alpha$

Recall: z_α is the value such that $P(Z > z_\alpha) = \alpha$, from the Standard Normal table.

Example 1: Test for μ, σ known

The nominal capacity of a soft drink bottle is 330 mL. The filling process has $\sigma = 5$ mL. A random sample of $n = 40$ bottles has $\bar{x} = 328.2$ mL. At $\alpha = 0.05$, test whether the mean capacity differs from 330 mL.

Example 1: Solution

Hypotheses: $H_0 : \mu = 330$ vs. $H_1 : \mu \neq 330$ (two-tailed)

Test statistic: $Z_0 = \frac{\bar{X}-330}{5/\sqrt{40}} \sim N(0,1)$

Observed value:

$$z_{obs} = \frac{328.2 - 330}{5/\sqrt{40}} = \frac{-1.8}{0.7906} \approx -2.28$$

Rejection region (two-tailed, $\alpha = 0.05$):

$$|z_{obs}| \geq z_{0.025} = 1.96$$

Since $|-2.28| = 2.28 > 1.96$, we **reject** H_0.

Case 2: Test for μ, σ Unknown

Test for μ — Normal population, σ unknown

Test statistic:

$$T_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}} \underset{\text{under } H_0}{\sim} t_{(n-1)}$$

Critical regions (at significance level α):

H_1	Rejection region
$\mu \neq \mu_0$	$\lVert t_{obs} \rVert \geq t_{\alpha/2,(n-1)}$
$\mu > \mu_0$	$t_{obs} \geq t_{\alpha,(n-1)}$
$\mu < \mu_0$	$t_{obs} \leq -t_{\alpha,(n-1)}$

Recall: S' is the corrected sample standard deviation, and $t_{\alpha,(\nu)}$ is from Table 7 (t-Student).

Example 2: Test for μ, σ unknown

An engineer suspects that the mean weight of packages produced by a machine exceeds the nominal value of 100 g. A random sample of $n = 20$ packages from a Normal population gives $\bar{x} = 101$ g and $s' = 3$ g.

Test at $\alpha = 0.05$.

Example 2: Solution

Hypotheses: $H_0 : \mu \leq 100$ vs. $H_1 : \mu > 100$ (right-tailed)

Test statistic: $T_0 = \frac{\bar{X}-100}{S'/\sqrt{20}} \sim t_{(19)}$

Observed value:

$$t_{obs} = \frac{101 - 100}{3/\sqrt{20}} = \frac{1}{0.6708} \approx 1.49$$

Rejection region (right-tailed, $\alpha = 0.05$):

$$t_{obs} \geq t_{0.05,(19)} = 1.729$$

(Table 7)

Case 2’: Test for μ, σ Unknown, $n > 30$

Test for μ — Normal population, σ unknown, large sample ($n > 30$)

When $n > 30$, S' approximates σ well and the t-distribution approaches the Standard Normal:

$$Z_0 = \frac{\bar{X} - \mu_0}{S'/\sqrt{n}} \underset{\text{under } H_0}{\dot{\sim}} N(0,1)$$

The critical regions are the same as in Case 1, using z_α and $z_{\alpha/2}$ from the Normal table.

> This is the same approach as in Case 1, but replacing σ by s'. The approximation is justified by the CLT and by the fact that $t_{(\nu)} \to N(0,1)$ as $\nu \to \infty$.

Case 3: Test for σ^2 (Variance)

Test for σ^2 — Normal population

Test statistic:

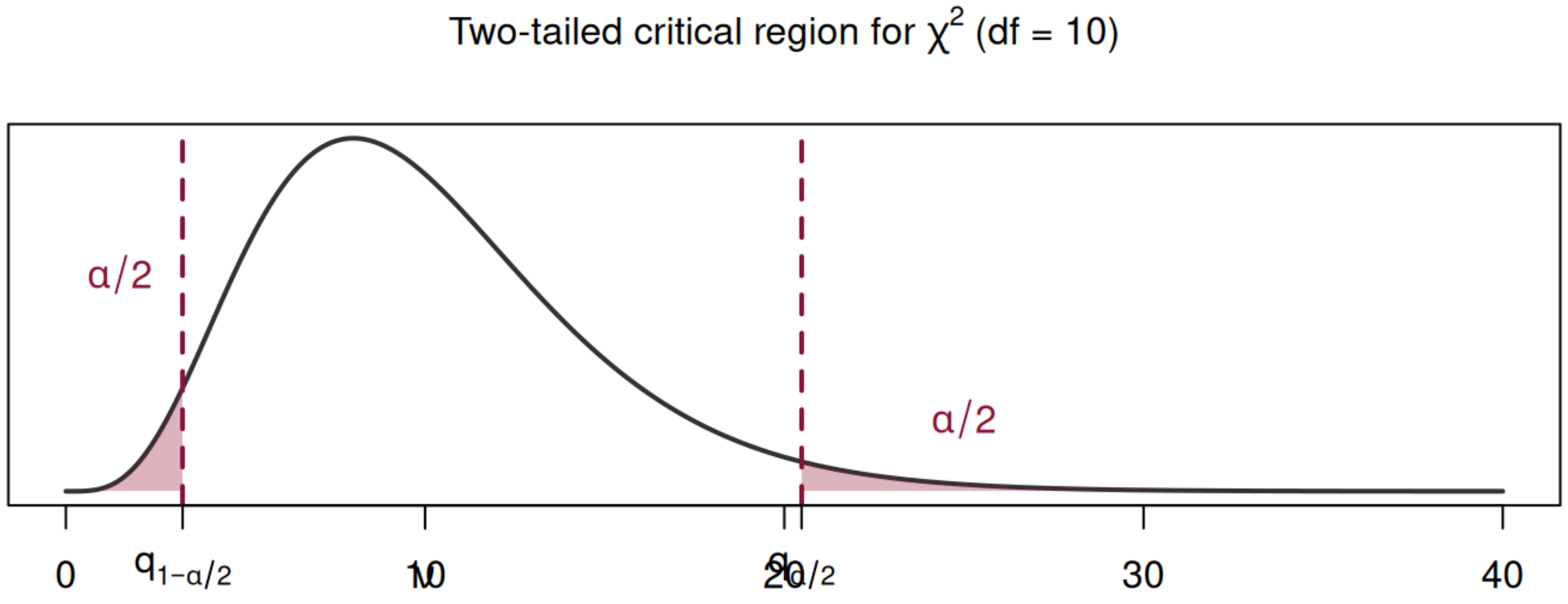
$$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2} \underset{\text{under } H_0}{\sim} \chi^2_{(n-1)}$$

Critical regions (at significance level α):

H_1	Rejection region
$\sigma^2 \neq \sigma_0^2$	$q_{obs} \leq q_{1-\alpha/2}$ or $q_{obs} \geq q_{\alpha/2}$
$\sigma^2 > \sigma_0^2$	$q_{obs} \geq q_{\alpha}$
$\sigma^2 < \sigma_0^2$	$q_{obs} \leq q_{1-\alpha}$

Here q_α denotes the value from the χ^2 table (Table 6) such that $P(Q > q_\alpha) = \alpha$.

Important: The χ^2 Distribution is Not Symmetric



Because the χ^2 distribution is **right-skewed**, the two-tailed rejection region is **not symmetric** around the center. The lower critical value is $q_{1-\alpha/2}$ and the upper is $q_{\alpha/2}$.

Example 3: Test for σ^2

Consider the test: $H_0 : \sigma^2 \geq 10$ vs. $H_1 : \sigma^2 < 10$, with $n = 20$ from a Normal population and $s'^2 = 9$ (i.e., $s' = 3$).

Test at $\alpha = 0.05$.

Example 3: Solution

Test statistic:

$$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2} \underset{\text{under } H_0}{\sim} \chi^2_{(19)}$$

Observed value:

$$q_{obs} = \frac{(20-1)\times 9}{10} = \frac{171}{10} = 17.1$$

Rejection region (left-tailed, $\alpha = 0.05$):

We need $q_{1-\alpha} = q_{0.95}$ from the χ^2 table with 19 d.f.

From Table 6: $q_{0.95,(19)} = 10.117$.

Summary of Cases — Section 3.3

Parameter	σ	Test Stat.	Distribution	Table
μ	known	$Z_0 = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$	$N(0,1)$	Normal
μ	unknown, $n \leq 30$	$T_0 = \frac{\bar{X}-\mu_0}{S'/\sqrt{n}}$	$t_{(n-1)}$	Table 7
μ	unknown, $n > 30$	$Z_0 = \frac{\bar{X}-\mu_0}{S'/\sqrt{n}}$	$\dot{\sim} N(0,1)$	Normal
σ^2	—	$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2}$	$\chi^2_{(n-1)}$	Table 6

This mirrors the cases from interval estimation — the pivot variables are the same.

Practice Exercise A

A quality inspector measures the temperature of a chemical process. Historical data show $\sigma = 2.5^\circ$C. A random sample of $n = 16$ measurements gives $\bar{x} = 98.3^\circ$C.

a. Test $H_0 : \mu = 100$ vs. $H_1 : \mu \neq 100$ at $\alpha = 0.01$.

b. Compute the p-value.

Practice Exercise A: Solution

a) $Z_0 = \frac{\bar{X}-100}{2.5/\sqrt{16}} \sim N(0,1)$

$$z_{obs} = \frac{98.3 - 100}{2.5/4} = \frac{-1.7}{0.625} = -2.72$$

Rejection region (two-tailed, $\alpha = 0.01$): $|z_{obs}| \geq z_{0.005} = 2.576$

Since $|-2.72| = 2.72 > 2.576$, we **reject** H_0.

b)
$p = 2 \times P(Z \geq 2.72) = 2 \times (1 - 0.9967) = 2 \times 0.0033 = 0.0066$

Conclusion: At 1%, there is evidence that the process temperature differs from 100°C.

Practice Exercise B

From a Normal population, a sample of $n = 15$ observations gives $s' = 5.04$.

Test $H_0 : \sigma^2 = 16$ vs. $H_1 : \sigma^2 \neq 16$ at $\alpha = 0.10$.

Practice Exercise B: Solution

Test statistic:

$$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2} \sim \chi^2_{(14)}$$

$$q_{obs} = \frac{14 \times (5.04)^2}{16} = \frac{14 \times 25.4016}{16} = \frac{355.62}{16} \approx 22.23$$

Rejection region (two-tailed, $\alpha = 0.10$):

From Table 6, $\nu = 14$: $q_{0.95,(14)} = 6.571$ and $q_{0.05,(14)} = 23.685$

Reject if $q_{obs} \leq 6.571$ or $q_{obs} \geq 23.685$.

Since $6.571 < 22.23 < 23.685$, we **do not reject** H_0.

Quick Recap Before the Break

What we covered (Section 3.3)

- **Case 1**: Test for μ with σ known $\rightarrow Z_0 \sim N(0,1)$
- **Case 2**: Test for μ with σ unknown $\rightarrow T_0 \sim t_{(n-1)}$
- **Case 2'**: Test for μ with σ unknown, $n > 30 \rightarrow Z_0 \dot{\sim} N(0,1)$
- **Case 3**: Test for $\sigma^2 \rightarrow Q_0 \sim \chi^2_{(n-1)}$

After the break: Sections 3.4 and 3.5.